

Student Number _____



2020

HSC Assessment 1 Term 4

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 80 minutes
- Board approved calculators may be used.
- Write using black or blue pen
- All necessary working should be shown in all questions
- Write your student number at the top of every page of your answer sheets.
- NESA Reference Sheet is provided

Total marks – 68

Attempt all questions

Start each question on a new sheet of paper.

NESA Reference Sheet is provided

This paper MUST NOT be removed from the examination room

Section A Multiple Choice (5 Marks)

QUESTION 1

It is known that $(x + 2)$ is a factor of the polynomial $P(x)$ and that

$$P(x) = (x^2 + x + 1) \times Q(x) + (2x + 3)$$

for some polynomial $Q(x)$. From this information alone, it can be deduced that:

(A) $Q(-2) = -\frac{1}{3}$

(B) $Q(-2) = \frac{1}{3}$

(C) $Q(2) = -1$

(D) $Q(2) = 1$

QUESTION 2

The roots of the polynomial $4x^3 + 4x - 5 = 0$ are α, β and γ .

What is the value of $(\beta + \gamma - 3\alpha)(\alpha + \beta - 3\gamma)(\gamma + \alpha - 3\beta)$?

(A) -80

(B) 80

(C) -16

(D) 16

QUESTION 3

If $\tan x = \frac{a}{b}$, then $b \cos 2x + a \sin 2x$ is equal to

(A) $\frac{a}{b}$

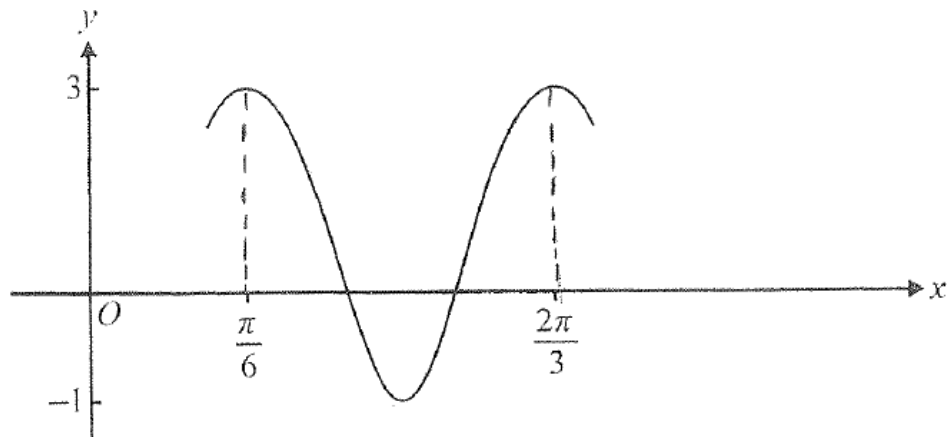
(B) $\frac{b}{a}$

(C) a

(D) b

QUESTION 4

The graph below could have the equation



- (A) $y = 1 + 2\cos(x - \frac{\pi}{6})$
- (B) $y = 1 + 2\cos(x + \frac{\pi}{6})$
- (C) $y = 1 + 2\cos 4(x - \frac{\pi}{6})$
- (D) $y = 1 + 2\cos 4(x + \frac{\pi}{6})$

QUESTION 5

The number of values of x in $[0, 2\pi]$ that satisfies the equation

$$(\sin x)^2 - \cos x = \frac{1}{4}$$

- (A) 1
- (B) 2
- (C) 3
- (D) 4

QUESTION 6 Start a new page**9 marks**

(a) Solve $4\cos x + 3\sin x = -4$ using the result $t = \tan \frac{x}{2}$ for $0^\circ \leq x \leq 360^\circ$ **3**

(b) Solve $\cos 2x + \cos x + 1 = 0$ for $0 \leq x \leq 2\pi$ **3**

(c) Prove that for all integers $n \geq 1$,

$$1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{4n^3 - n}{3} \quad \mathbf{3}$$

QUESTION 7 Start a new page**9 marks**

(a) The numbers α, β and γ satisfy the equations

$$\alpha + \beta + \gamma = 3$$

$$\alpha^2 + \beta^2 + \gamma^2 = 1$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = 2.$$

(i) Find the values of $\alpha\beta + \beta\gamma + \gamma\alpha$ and $\alpha\beta\gamma$. **2**

(ii) Explain why α, β and γ are the roots of the cubic equation **1**

$$x^3 - 3x^2 + 4x - 2 = 0.$$

(b) When $-2x^2 + ax + 4$ is divided by $x - 1$ and $2x + 1$ respectively, the remainders are equal. Find the value of a . **3**

(c) Use the substitution $t = \tan \frac{x}{2}$ to show that

$$\sec x + \tan x = \tan\left(\frac{\pi}{4} + \frac{x}{2}\right) \quad \mathbf{3}$$

QUESTION 8 Start a new page**9 marks**

- (a) Graph the polynomial function $P(x) = x^3 + 2x^2 - 5x - 6$ **3**
- (b) Given that
$$5 \cos \theta - 5\sqrt{3} \sin \theta = A \cos(\theta + \alpha),$$
- (i) Find values of A and α where $A > 0$ and $0 < \alpha < \frac{\pi}{2}$ **2**
- (ii) Hence find the maximum value of $5 \cos \theta - 5\sqrt{3} \sin \theta + 15$ **1**
- (c) Prove by mathematical induction that for all integers $n \geq 0$
 $3^{4n} - 1$ is divisible by 80 **3**

QUESTION 9 Start a new page**9 marks**

- (a) The polynomial $P(x) = x^4 + 7x^3 + 9x^2 - 27x + C$ has a triple zero.
- (i) Determine the value of the triple zero **2**
- (ii) Hence, find the value of C **1**
- (iii) Factorise $P(x)$ **1**
- (iv) Sketch the graph of $y = P(x)$ **1**
- (b) (i) Sketch the curve $y = \frac{1}{2} \sin(2x - \pi) + 1$ for $0 \leq x \leq 2\pi$ **2**
- (ii) Hence or otherwise find all the solutions of $\frac{1}{2} \sin(2x - \pi) = -\frac{1}{2}$ in an unrestricted domain **2**

(a) (i) Prove $\sin 3x = 3\sin x - 4\sin^3 x$ 2

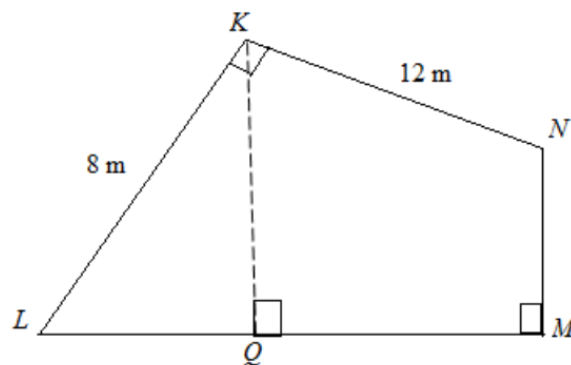
(ii) Hence or otherwise solve $3\sin x - 4\sin^3 x = \frac{\sqrt{3}}{2}$ in the domain $[0, 2\pi]$ 2

(b) The diagram shows a pond KLMN in a park.

LK = 8 metres and KN = 12 metres.

$\angle LKN = \angle LMN = 90^\circ$ and $\angle KLM = \theta$, where $0^\circ < \theta < 90^\circ$.

The perimeter of the pond is P metres and QK is perpendicular to LM.



(i) Show that the perimeter KLMN can be written in the form $P = a + b \cos \theta + d \sin \theta$ where a, b , and d are 20, -4 and 20 respectively 2

(ii) Express P in the form of $a + R \sin (\theta - \beta)$,
Where $R > 0$ and $0^\circ < \beta < 90^\circ$, β to nearest minute 2

(iii) Hence, find the value of θ (to nearest minute)
when $P=38$ metres 1

QUESTION 11 Start a new page**9 marks**

(a) A polynomial of degree n is given by $P(x) = x^n + ax - b$.

It is given that the polynomial has a double root at $x = a$.

Find the derived polynomial $P'(x)$ and show that $a^{n-1} = -\frac{a}{n}$ **2**

(b) Given that $81^{(\sin^2 x)} + 81^{(\cos^2 x)} = 30$

(i) Show that the above equation can be written in the form

$$y^2 - 30y + 81 = 0 \text{ where } y = 81\sin^2 x \quad \mathbf{1}$$

(ii) Solve $y^2 - 30y + 81 = 0$ **1**

(iii) Hence or otherwise, find all solutions of the equation

$$81^{(\sin^2 x)} + 81^{(\cos^2 x)} = 30, \text{ where } (0, \pi] \quad \mathbf{2}$$

(c) Use Mathematical Induction to prove for integers $n \geq 1$

$$1(1!) + 2(2!) + 3(3!) + \dots + n(n!) = (n+1)! - 1 \quad \mathbf{3}$$

QUESTION 12 Start a new page**9 marks**

a) Given the identity $\sin 5\theta = 16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta$.

(i) Deduce that $x = \sin\left(\frac{\pi}{10}\right)$ is one of the solutions to

$$16x^5 - 20x^3 + 5x - 1 = 0 \quad \mathbf{2}$$

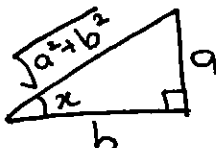
(ii) Find the polynomial $P(x)$ such that

$$(x-1)P(x) = 16x^5 - 20x^3 + 5x - 1 \quad \mathbf{2}$$

(iii) Find the value of a such that $P(x) = (4x^2 + ax - 1)^2$ **1**

(iv) Hence, find an exact value for $\sin\left(\frac{\pi}{10}\right)$ **1**

(b) Prove by Mathematical Induction that $a^n - b^n$ is divisible by $a - b$ for all positive integers n . **3**

MATHEMATICS Extension 1 : Question MC..		
Suggested Solutions	Marks	Marker's Comments
1. $P(x) = (x^2 + x + 1)Q(x) + (2x + 3)$ $P(-2) = ((-2)^2 + (-2) + 1)Q(-2) + (2(-2) + 3)$ $0 = 3Q(-2) - 1$ $3Q(-2) = 1$ $Q(-2) = \frac{1}{3}$ (B) 2. $\alpha + \beta + \gamma = 0$ $\alpha\beta + \alpha\gamma + \beta\gamma = 1$ $\alpha\beta\gamma = \frac{5}{4}$ $(\beta + \gamma - 3\alpha)(\alpha + \beta - 3\gamma)(\gamma + \alpha - 3\beta)$ $= (\beta + \gamma + \alpha - 4\alpha)(\alpha + \beta + \gamma - 4\gamma)(\gamma + \alpha + \beta - 4\beta)$ $= (0 - 4\alpha)(0 - 4\gamma)(0 - 4\beta)$ $= -64\alpha\beta\gamma$ $= -64\left(\frac{5}{4}\right)$ $= -80$ (A) 3. $\tan x = \frac{a}{b}$  $b\cos 2x + a\sin 2x$ $= b(\cos^2 x - \sin^2 x) + 2a\sin x \cos x$ $= b\left(\frac{b^2}{a^2 + b^2} - \frac{a^2}{a^2 + b^2}\right) + 2a\left(\frac{a}{\sqrt{a^2 + b^2}}\right)\left(\frac{b}{\sqrt{a^2 + b^2}}\right)$ $= \frac{b^3 - a^2b}{a^2 + b^2} + \frac{2a^2b}{a^2 + b^2}$ $= \frac{b^3 + a^2b}{a^2 + b^2}$ $= \frac{b(b^2 + a^2)}{a^2 + b^2}$ $= b$ (D)		

MATHEMATICS Extension 1 : Question .. M.C

Suggested Solutions	Marks	Marker's Comments
4. Period = $\frac{\pi}{2}$ $\therefore \frac{2\pi}{n} = \frac{\pi}{2}$ $n = 4$ (C) 5. $\sin^2 x - \cos x = \frac{1}{4}$ $1 - \cos^2 x - \cos x = \frac{1}{4}$ $4 - 4\cos^2 x - 4\cos x = 1$ $4\cos^2 x + 4\cos x - 3 = 0$ $\cos x = \frac{-4 \pm \sqrt{16 + 4(4)(3)}}{2 \times 4}$ $= \frac{1}{2}, -\frac{3}{2}$ $\cos x = \frac{1}{2}$ $\cos x = -\frac{3}{2}$ $x = \frac{\pi}{3}, \frac{5\pi}{3}$ no solution. $\therefore$ 2 solutions (B)		

$$\frac{4(1-t^2)}{1+t^2} + \frac{6t}{1+t^2} = -4$$

$$4 - \cancel{4t^2} + 6t = -4 - \cancel{4t^2}$$

$$\textcircled{1} \quad \begin{aligned} 6t &= -8 \\ t &= -\frac{4}{3} \end{aligned}$$

$$\tan \frac{x}{2} = -\frac{4}{3} \quad \text{for } 0 \leq \frac{x}{2} \leq 180^\circ$$

$\textcircled{-1}$ — Domain.

$$\frac{x}{2} = 180 - 53^\circ 7' 48.31''$$

$$\frac{x}{2} = 126^\circ 52' 11.6''$$

$$\textcircled{i} \quad x = 253^\circ 44' \text{ (nearest minute)}$$

check 180° :

$$\text{LHS} = 4(-1) + 3(0) = -4 = \text{RHS so } 180^\circ \text{ is a}$$

solution

$$\text{So } x = 180^\circ, 253^\circ 44' \text{ (nearest minute).}$$

$$253.74^\circ \text{ (2dp)}$$

(b) $\cos 2x + \cos x + 1 = 0$

① $\cos^2 x - \sin^2 x + \cos x + 1 = 0$

$\cos^2 x - (1 - \cos^2 x) + \cos x + 1 = 0$

$2\cos^2 x + \cos x = 0$

$\cos x (2\cos x + 1) = 0$

$\cos x = 0$ or $2\cos x + 1 = 0$

$x = \frac{\pi}{2}, \frac{3\pi}{2}$

$\cos x = -\frac{1}{2}$

$x = \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}$

$= \frac{2\pi}{3}, \frac{4\pi}{3}$

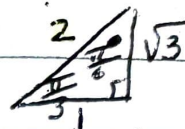
∴ $x = \frac{\pi}{2}$

$\frac{2\pi}{3}, \frac{4\pi}{3}, \frac{3\pi}{2}$

$\frac{\pi}{2}, \frac{3\pi}{2}, \frac{2\pi}{3}, \frac{4\pi}{3}$

$0 \leq x \leq 2\pi$

$(2\cos^2 x - 1) + (\cos x + 1) = 0$



(4) RTP $1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{4n^3 - n}{3}$

n=1 LHS = $\frac{(2(1)-1)^2}{1} = 1$

RHS = $\frac{4(1)^3 - (1)}{3} = \frac{3}{3} = 1$

LHS = RHS

Hence true for n=1

show substitution (1)

Assume true for n=k & prove true for n=k+1

n=k $1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 = \frac{4k^3 - k}{3}$

n=k+1

RTP $1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 + (2(k+1)-1)^2 = \frac{4(k+1)^3 - (k+1)}{3}$

LHS = $\frac{4k^3 - k}{3} + (2(k+1)-1)^2$

RHS = $\frac{4(k^3 + 3k^2 + 3k + 1) - k - 1}{3}$

Correct assumption (1)

$= \frac{4k^3 - k}{3} + (2k+1)^2$
 $= \frac{4k^3 - k + 3(4k^2 + 4k + 1)}{3}$

$= \frac{4k^3 + 12k^2 + 11k + 3}{3}$

$= 4k^3 - k + 12k^2 + 12k + 3$

$= \frac{4k^3 + 12k^2 + 11k + 3}{3}$

show all steps

$= \frac{4k^3 + 12k^2 + 11k + 3}{3}$

= RHS

Since true for n=1 it is true for n=1+1 ie n=2 & so on for all positive integral n ≥ 1

MATHEMATICS Extension 1 : Question ...7...

Suggested Solutions	Marks	Marker's Comments
a) i) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$ $1 = 3^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$ $\therefore \alpha\beta + \alpha\gamma + \beta\gamma = \frac{3^2 - 1}{2}$ $= 4$ $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = 2$ $\frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma} = 2$ $\therefore \alpha\beta\gamma = \frac{4}{2}$ $= 2$ $\therefore \alpha\beta + \alpha\gamma + \beta\gamma = 4$ $\alpha\beta\gamma = 2$ ii) $x^3 - 3x^2 + 4x - 2 = 0$ let the roots be a, b, c sum of roots $a + b + c = 3$ from (i) $\alpha + \beta + \gamma = 3$ sum of paired roots $ab + ac + bc = 4$ from (i) $\alpha\beta + \alpha\gamma + \beta\gamma = 4$ product of roots $abc = 2$ from (i) $\alpha\beta\gamma = 2$ $\therefore \alpha = a, \beta = b$ and $\gamma = c$ $\therefore \alpha, \beta$ and γ are the roots of $x^3 - 3x^2 + 4x - 2 = 0$	1 1 1 1	must have some explanation and show $\sum \alpha, \sum \alpha\beta$ and $\alpha\beta\gamma$.
b) let $P(x) = -2x^2 + ax + 4$ $P(1) = -2(1)^2 + a(1) + 4$ $= 2 + a$ $P(-\frac{1}{2}) = -2(-\frac{1}{2})^2 + a(-\frac{1}{2}) + 4$ $= \frac{7}{2} - \frac{a}{2}$ $P(1) = P(-\frac{1}{2})$ $2 + a = \frac{7}{2} - \frac{a}{2}$ $\frac{3}{2}a = \frac{3}{2}$	1 1	
$\therefore a = 1$	1	

MATHEMATICS Extension 1 : Question ...7...

Suggested Solutions	Marks	Marker's Comments
c) $\sec x + \tan x = \tan\left(\frac{\pi}{4} + \frac{x}{2}\right)$ LHS = $\sec x + \tan x$ $= \frac{1+t^2}{1-t^2} + \frac{2t}{1-t^2}$ $= \frac{t^2 + 2t + 1}{1-t^2}$ $= \frac{(t+1)^2}{(1-t)(1+t)}$ $= \frac{t+1}{1-t}$ $= \frac{1+t}{1-1 \times t}$ $= \frac{\tan \frac{\pi}{4} + \tan \frac{x}{2}}{1 - \tan \frac{\pi}{4} \tan \frac{x}{2}}$ $= \tan\left(\frac{\pi}{4} + \frac{x}{2}\right)$ = RHS	1 1 1	for recognising that $\tan \frac{\pi}{4} = 1$ and writing the compound angle correctly.

MATHS EXTENSION I

HSC Assessment 1 – Term
4 (2020)

Questions 8

(a) Graph the polynomial function $P(x) = x^3 + 2x^2 - 5x - 6$

(b) Given that: $5\cos\theta - 5\sqrt{3}\sin\theta = A\cos(\theta + \alpha)$,

i. Find values of A and α where $A > 0$ and $0 < \alpha < \frac{\pi}{2}$

ii. Hence find the maximum value of $5\cos\theta - 5\sqrt{3}\sin\theta + 15$

(c) Prove by mathematical induction that for all integers $n \geq 0$, $3^{4n} - 1$ is divisible by 80

Q8(a)

Find x-intercepts

$$P(x) = x^3 + 2x^2 - 5x - 6$$

$$P(1) = 1 + 2 - 5 - 6$$

$$P(-1) = -1 + 2 + 5 - 6$$

$\therefore x + 1$ is a factor

$$P(x) = (x + 1)(x^2 + x - 6)$$

$$= (x + 1)(x + 3)(x - 2)$$

$\therefore x - \text{intercepts are } -1, -3 \text{ and } 2$

Q8(a)

Find y-intercept

y-intercept @ $x = 0$

$$P(0) = 0^3 + 2(0)^2 - 5(0) - 6$$

$$P(0) = -6$$

$\therefore y - \text{intercept is } -6$

1

Finding additional points

$$P\left(\frac{1}{2}\right) = -\frac{63}{8}$$

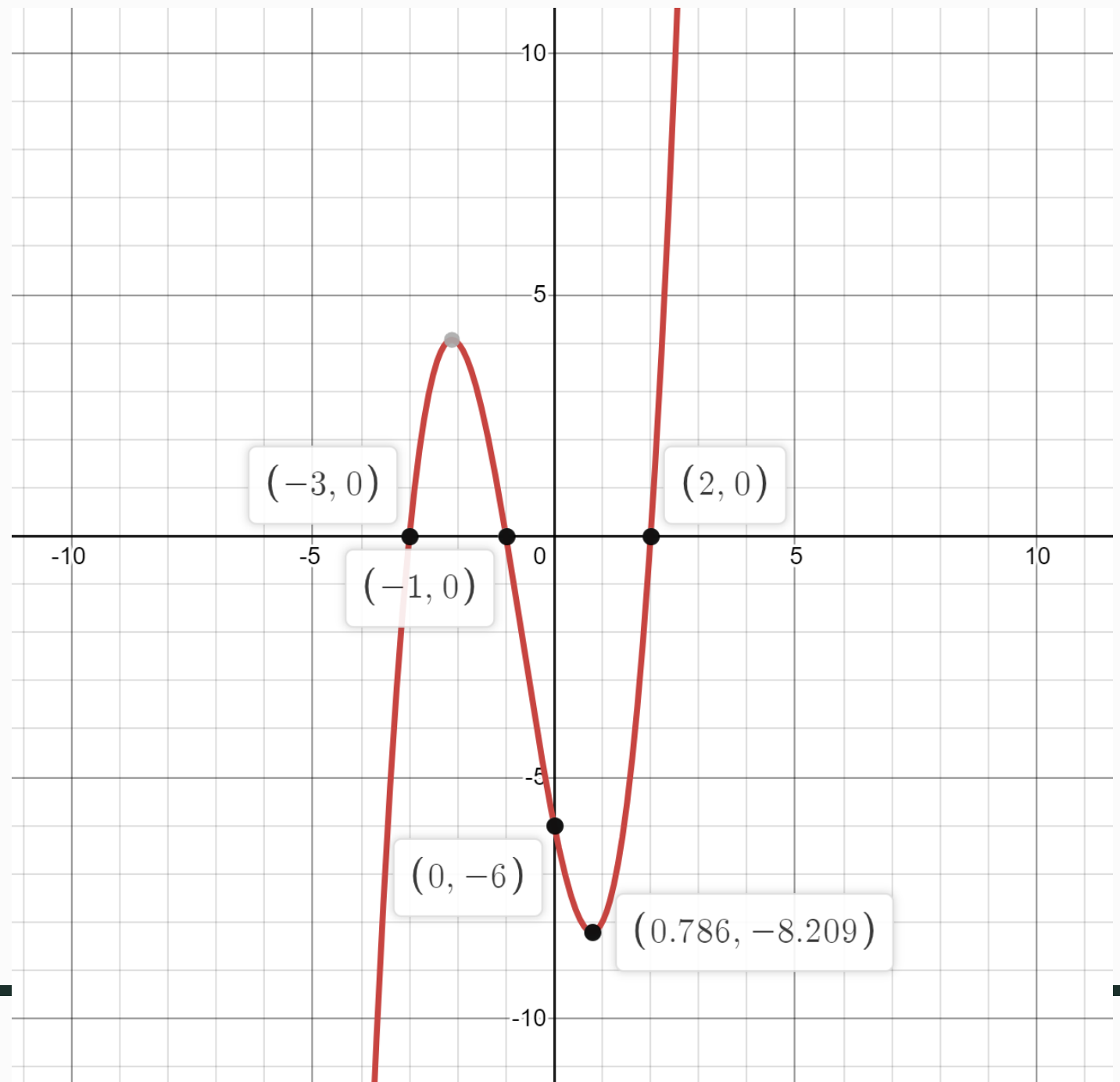
$$P(-2) = 4$$

Graph

1 x-intercepts

1 y-intercepts

1 shape



Q8(b)(i)

$$5\cos\theta - 5\sqrt{3}\sin\theta = A\cos(\theta + \alpha)$$

$$5\cos\theta - 5\sqrt{3}\sin\theta = A\cos\theta\cos\alpha - A\sin\theta\sin\alpha$$

$$5 = A\cos\alpha$$

$$\cos\alpha = \frac{5}{A}$$

$$-5\sqrt{3} = -A\sin\alpha$$

$$\sin\alpha = \frac{5\sqrt{3}}{A}$$

$$A = \sqrt{5^2 + (5\sqrt{3})^2}$$

$$= \sqrt{100}$$

$$= 10$$

Since $A > 0$

1

$$\tan\alpha = \frac{5\sqrt{3}}{5}$$

$$= \sqrt{3}$$

$$\alpha = \frac{\pi}{3}$$

Since $0 < \alpha < \frac{\pi}{2}$

1

Q8(b)(ii)

$$5\cos\theta - 5\sqrt{3}\sin\theta = 10\cos\left(\theta + \frac{\pi}{3}\right) + 15$$

$$\therefore \text{maximum value} = 10 + 15$$

$$= 25$$

1

Q8(c)

Prove $3^{4n} - 1$ is divisible by 80.

Test $n = 0$

$$\begin{aligned} 3^{4n} - 1 &= 3^{4(0)} - 1 \\ &= 1 - 1 \\ &= 0 \text{ (which is divisible by 80)} \end{aligned}$$

1

True for $n = 0$

Assume true for $n = k$ (where k is some positive integer):

$$3^{4k} - 1 = 80M \text{ (M is an integer)}$$

Prove true for $n = k + 1$:

$$\begin{aligned} 3^{4(k+1)} - 1 &= 3^{4k+4} - 1 \\ &= 3^{4k} \times 3^4 - 1 \\ &= 3^{4k} \times 81 - 1 \\ &= (3^{4k} - 1) \times 81 + 80 \\ &= 80M \times 81 + 80 \\ &= 80(81M + 1) \end{aligned}$$

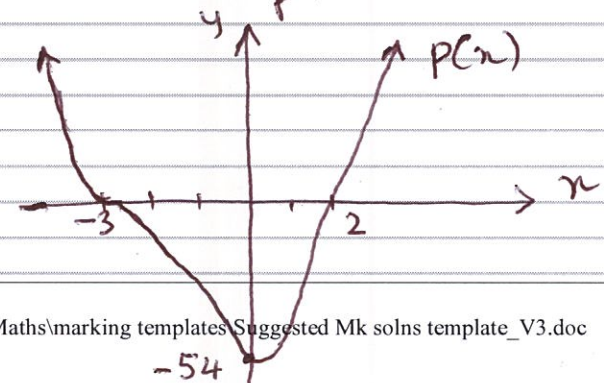
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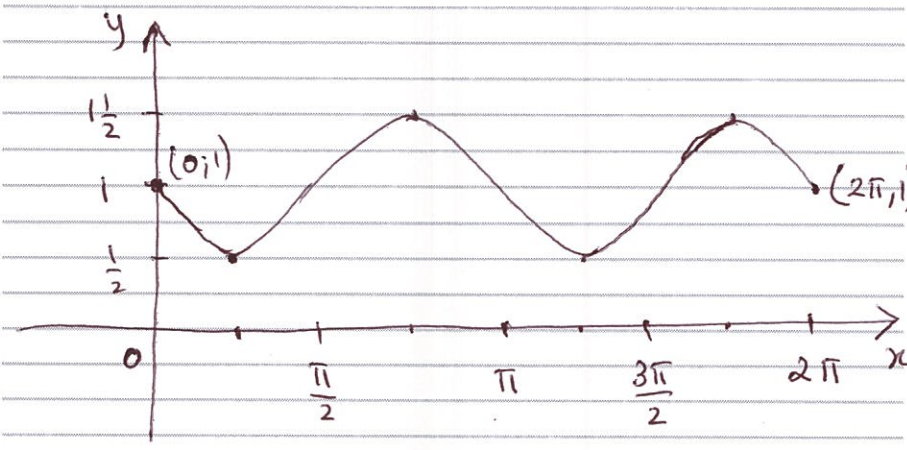
Which is divisible by 80.
True for $n = k + 1$

Since true for $n = 0$, by PMI, true for any integral ≥ 0 .

MATHEMATICS Extension 1 : Question.....9

Suggested Solutions	Marks	Marker's Comments
<u>Question 9:</u> (a) $p(x) = x^4 + 7x^3 + 9x^2 - 27x + C$ (i) $p(x)$ has a triple zero let α is the triple zero $\therefore p(\alpha) = p'(\alpha) = p''(\alpha) = 0$ $p'(x) = 4x^3 + 21x^2 + 18x - 27$ $p''(x) = 12x^2 + 42x + 18$ $= 6(2x+1)(x+3)$ $p''(\alpha) = 0 \therefore 6(2\alpha+1)(\alpha+3) = 0$ $\therefore \alpha = -\frac{1}{2}$ or $\alpha = -3$ $p'(-\frac{1}{2}) = 4(-\frac{1}{2})^3 + 21(-\frac{1}{2})^2 + 18(-\frac{1}{2}) - 27$ $= -\frac{125}{4} \neq 0$ $\therefore -\frac{1}{2}$ is not a triple zero $p'(-3) = 4(-3)^3 + 21(-3)^2 + 18(-3) - 27$ $= 0 \therefore \alpha = -3$ is the triple zero (ii) $p(-3) = 0 \therefore (-3)^4 + 7(-3)^3 + 9(-3)^2 - 27(-3) + C = 0$ $C = -54$ (iii) $p(x) = (x+3)^3(x-2)$ Since sum of roots = -7 (iv) 	1 1 1 1	need to show all working to get the mark.

MATHEMATICS Extension 1 : Question.....

Suggested Solutions	Marks	Marker's Comments
(b) (i) $y = \frac{1}{2} \sin(2x - \pi) + 1$  (ii) $\frac{1}{2} \sin(2x - \pi) = -\frac{1}{2}$ $\sin(2x - \pi) = -1 = \sin\left(-\frac{\pi}{2}\right)$ $2x - \pi = -\frac{\pi}{2} + 2k\pi \quad k \in \mathbb{Z}$ $2x = \frac{\pi}{2} + 2k\pi$ $x = \frac{\pi}{4} + k\pi \quad k \in \mathbb{Z}$ Graphically, looking for solutions when $y = \frac{1}{2}$ $\therefore x = \dots, -\frac{11\pi}{4}, -\frac{7\pi}{4}, -\frac{3\pi}{4}, \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4}, \dots$	2 2	1 for shape 1 for max/min and y-intercept 1 mark for some solutions (at least one) correct The two marks only for full correct solutions.

a) (i) $\sin 3x = \sin(2x+x)$

$$\begin{aligned} &= \sin 2x \cos x + \cos 2x \sin x \\ &= 2\sin x \cos^2 x + (1-2\sin^2 x)\sin x \\ &= 2\sin x(1-\sin^2 x) + \sin x - 2\sin^3 x \\ &= 2\sin x - 2\sin^3 x + \sin x - 2\sin^3 x \\ &= 3\sin x - 4\sin^3 x \end{aligned}$$

- ① expansion and double angle
① simplification involving Pythagorean identity

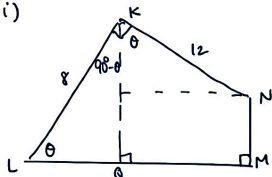
(ii) $\sin 3x = \frac{\sqrt{3}}{2} \quad 0 \leq 3x \leq 6\pi$

$$3x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{7\pi}{3}, \frac{8\pi}{3}, \frac{13\pi}{3}, \frac{14\pi}{3}$$

$$x = \frac{\pi}{9}, \frac{2\pi}{9}, \frac{7\pi}{9}, \frac{8\pi}{9}, \frac{13\pi}{9}, \frac{14\pi}{9}$$

- ① 4 solutions
① 6 solutions

b) (i)



Students should not introduce pronounal without showing or explaining very clearly where it is.

$$\cos \theta = \frac{LQ}{8}$$

$$LQ = 8 \cos \theta$$

$$\sin \theta = \frac{QM}{12}$$

$$QM = 12 \sin \theta$$

- ① LQ, QM are in each triangle

$$NM = 8 \sin \theta - 12 \cos \theta$$

- ① NM + P

$$P = 8 + 12 + 8 \sin \theta - 12 \cos \theta + 12 \sin \theta + 8 \cos \theta$$

$$= 20 + 20 \sin \theta - 4 \cos \theta$$

$$= 20 - 4 \cos \theta + 20 \sin \theta$$

(ii) $R \sin(\theta - \beta) = R \sin \theta \cos \beta - R \cos \theta \sin \beta = 20 \sin \theta - 4 \cos \theta$

$$R \cos \beta = 20$$

$$R \sin \beta = 4$$

$$R = \sqrt{16 + 400}$$

$$= \sqrt{416}$$

$$\beta = \tan^{-1} \frac{1}{5}$$

$$= 11.0191^\circ$$

$$\therefore P = 20 + \sqrt{416} \sin(\theta - 11.0191^\circ)$$

- ① R & β

- ① P

(iii)

$$\sqrt{416} \sin (\theta - 11^{\circ} 19') = 18$$
$$\sin (\theta - 11^{\circ} 19') = \frac{18}{\sqrt{416}}$$
$$\theta - 11^{\circ} 19' = 61^{\circ} 57'$$

$$\theta = 73^{\circ} 16'$$

①

(or $73^{\circ} 15'$ using exact values)

MATHEMATICS Extension 1 : Question 11...

Suggested Solutions

Marks

Marker's Comments

a) $P(x) = x^n + ax - b$

$$P'(x) = nx^{n-1} + a$$

(1)

$$P'(a) = 0 \quad (\text{Since } x=a \text{ is a double root})$$

$$\therefore na^{n-1} + a = 0$$

$$na^{n-1} = -a$$

$$a^{n-1} = \frac{-a}{n}$$

(1)

b) i $y = 81^{\sin^2 x}$

$$y = \frac{81^{\cos^2 x}}{1 - \sin^2 x} = 81$$

$$= \frac{81}{81^{\sin^2 x}}$$

$$\therefore 81^{\sin^2 x} + \frac{81}{81^{\sin^2 x}} = 30$$

$$(81^{\sin^2 x})^2 + 81 = 30 \times 81^{\sin^2 x}$$

let $y = 81^{\sin^2 x}$

$$\therefore y^2 + 81 = 30y$$

$$y^2 - 30y + 81 = 0$$

(1)

MATHEMATICS Extension 1 : Question...ii...

Suggested Solutions

Marks

Marker's Comments

ii) $y^2 - 30y + 81 = 0$

$$(y - 27)(y - 3) = 0$$

$$y = 27 \text{ or } 3$$

①

iii) Solve $81^{\sin^2 x} = 27 \text{ or } 3.$

$$\therefore \sin^2 x = \frac{3}{4} \text{ or } \frac{1}{4}$$

$$\therefore \sin x = \frac{\pm\sqrt{3}}{2} \text{ or } \pm \frac{1}{2}$$

①

$$\therefore x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{3}, \frac{2\pi}{3}.$$

①

MATHEMATICS Extension 1 : Question...!!...

Suggested Solutions

Marks

Marker's Comments

c) Test for $n=1$,

$$\begin{aligned} \text{LHS} &= 1(1!) \\ &= 1 \end{aligned} \quad \begin{aligned} \text{RHS} &= (1+1)! - 1 \\ &= 2 - 1 \\ &= 1 \end{aligned}$$

$$\therefore \text{LHS} = \text{RHS}$$

$\therefore$ The statement is true for $n=1$

(1)

Assume the statement is true for $n=k$, $k \in \mathbb{Z}^+$

$$\therefore 1(1!) + 2(2!) + \dots + k(k!) = (k+1)! - 1$$

Test for $n=k+1$,

$$\text{RTP: } 1(1!) + 2(2!) + 3(3!) + \dots + (k+1)[(k+1)!] = (k+2)! - 1$$

$$\text{LHS} = [1(1!) + 2(2!) + 3(3!) + \dots + k(k!)] + (k+1)[(k+1)!]$$

$$= [(k+1)! - 1] + (k+1)[(k+1)!] \quad (\text{By assumption})$$

$$= (k+1)! [1 + (k+1)] - 1$$

$$= (k+1)! (k+2) - 1$$

$$= (k+2)!$$

$\therefore$ If the statement is true for $n=k$, it is true for $n=k+1$

$\therefore$ The statement is true for all positive integer n by the principle of mathematical induction.

(1)

Mathematics Extension 1: Question 12

Solution	Marks	Marker's Comments
a) i) <u>Correct Method</u> $16x^5 - 20x^3 + 5x - 1 = 0$ Let $x = \sin\theta$ $\therefore 16\sin^5\theta - 20\sin^3\theta + 5\sin\theta - 1 = 0$ $\therefore \sin 5\theta - 1 = 0 \quad (\text{given } \sin 5\theta = 16\sin^5\theta - 20\sin^3\theta + 5\sin\theta)$ $\therefore \sin 5\theta = 1$ $5\theta = \frac{\pi}{2} + 2k\pi \quad (k \in \mathbb{N})$ $\theta = \frac{\pi}{10} + \frac{2}{5}k\pi$ When $k = 0$, $\theta = \frac{\pi}{10}$ $\therefore x = \sin \frac{\pi}{10} \text{ is one of the solutions}$ <u>Incomplete Method</u> When $x = \sin \frac{\pi}{10}$ $16x^5 - 20x^3 + 5x - 1$ $= 16\sin^5 \frac{\pi}{10} - 20\sin^3 \frac{\pi}{10} + 5\sin \frac{\pi}{10} - 1$ $= \sin(5 \times \frac{\pi}{10}) - 1 \quad (\text{given } \sin 5\theta = 16\sin^5\theta - 20\sin^3\theta + 5\sin\theta)$ $= \sin \frac{\pi}{2} - 1 = 0$	Total=2 1 1	The question asks to prove that $x = \sin \frac{\pi}{10}$ is one of the solutions. It means: ① $x = \sin \frac{\pi}{10}$ is a solution ② there are more than one solution Many students forget to add $2k\pi$ to the solution, which means they fail to prove ②. Thus, they lose 1 mark. This method can only receive up to 1 mark because it fails to prove ②.

[illegible]

Mathematics Extension 1: Question 12

Solution	Marks	Marker's Comments
a) ii) Method 2 - inspection By inspection, $P(x) = 16x^4 + 16x^3 - 4x^2 - 4x + 1$	2	Potential Mark: If "By inspection" or a similar statement is missing but the $P(x)$ expression is correct — 1 mark - Any answer without showing working out or mentioning "inspection method" but has the correct expression only receives 1 mark.
Method 3 - factor theorem From (i), the roots of the equation are $x = \sin \frac{\pi}{10}, \sin \frac{5\pi}{10}, \sin \frac{9\pi}{10}, \sin \frac{13\pi}{10}, \sin \frac{17\pi}{10}$ $\therefore 16x^5 - 20x^3 + 5x - 1$ $= 16(x - \sin \frac{\pi}{10})(x - \sin \frac{5\pi}{10})(x - \sin \frac{9\pi}{10})(x - \sin \frac{13\pi}{10})(x - \sin \frac{17\pi}{10})$ $= 16(x - \sin \frac{\pi}{10})(x - 1)(x - \sin \frac{9\pi}{10})(x - \sin \frac{13\pi}{10})(x - \sin \frac{17\pi}{10})$ $P(x) = 16(x - \sin \frac{\pi}{10})(x - \sin \frac{9\pi}{10})(x - \sin \frac{13\pi}{10})(x - \sin \frac{17\pi}{10})$	1	A few students use this method. Some forget to include the leading coefficient 16 in the factor form, thus receive 0 mark. In order to receive 2 marks for this method, the final answer needs to be in expanded form.

Mathematics Extension 1: Question 12

Solution	Marks	Marker's Comments
a) iii) $P(x) = (4x + ax - 1)^2$ $= 16x^4 + 16x^3 - 4x^2 - 4x + 1$ $(4x^2 + ax + 1)^2$ $= 16x^4 + 4ax^3 - 4x^2 + 4ax^3 + a^2x^2 - ax - 4x^2 - ax + 1$ Method 1: equate the coefficients x term $= -ax - ax = -2ax = -4x$ $a = 2$ Method 2: equate the coefficients x^3 term $= 4ax^3 + 4ax^3 = 8ax^3 = 16x^3$ $a = 2$	Total = 1	Note: the x or x^3 term has the simplest expression of a. It is a linear expression which avoid bringing additional solution like quadratic term does. x^2 term: $-4x^2 + a^2x^2 - 4x^2 = -4x^2$ $a^2 = 4$ $a = \pm 2$ Some students use the substitution method by subbing different values of x into the equation. Because $(4x^2 + ax - 1)^2$ has a square sign for the expression of a, it brings in an additional answer.

Mathematics Extension 1: Question 12

Solution	Marks	Marker's Comments
a) iv) $P(x) = 16x^5 - 20x^3 + 5x - 1$ $= (x-1)(4x^2 + 2x - 1)^2 = 0$ $\therefore x-1=0$ or $4x^2 + 2x - 1 = 0$ $x=1 \quad \text{or} \quad x = \frac{-2 \pm \sqrt{2^2 + 4 \times 4}}{8}$ $= \frac{-2 \pm \sqrt{20}}{8}$ $= \frac{-2 \pm 2\sqrt{5}}{8}$ $= \frac{-1 \pm \sqrt{5}}{4}$ $\therefore \sin \frac{7}{10} > 0$ $\therefore \sin \frac{7}{10} = \frac{1 + \sqrt{5}}{4}$	Total=1 1	Most students do well in this question. Common mistake: $\frac{-2 \pm 2\sqrt{5}}{8} = \frac{-1 \pm \sqrt{5}}{8}$

Mathematics Extension 1: Question 12

Solution	Marks	Marker's Comments
b) let $n=1$ $a^n - b^n = a - b = 1 \times (a - b)$ $1 \in \mathbb{N}$, therefore, the statement is true when $n=1$ Assume the statement is true for $n=k$ ($k \in \mathbb{N}^*$), this is, $a^k - b^k = p(a - b) \quad (p \in \mathbb{N})$ $a^k = b^k + p(a - b) \quad \textcircled{1} \quad \text{or}$ $b^k = a^k - p(a - b) \quad \textcircled{2}$ When $n=k+1$ $a^{k+1} - b^{k+1} = a^k \times a - b^k \times b$ (from $\textcircled{1}$) $= [b^k + p(a - b)] \times a - b^k \times b$ $= b^k \times a + ap(a - b) - b^k \times b$ $= b^k(a - b) + ap(a - b)$ $= (b^k + ap)(a - b)$ $\therefore b^k + ap \in \mathbb{N} \therefore$ the statement is true when $n=k+1$ or (from $\textcircled{2}$) $= a^k \times a - [a^k - p(a - b)] \times b$ $= a^k \times a - a^k \times b + bp(a - b)$ $= a^k(a - b) + bp(a - b)$ $= (a^k + bp)(a - b)$ $\therefore a^k + bp \in \mathbb{N} \therefore$ the statement is true when $n=k+1$ $\therefore$ By mathematical induction, the statement is true for all positive integers n	Total=3 1 1 1	The question should have restricted a and b to integers only. "A is divisible by B ($A, B \in \mathbb{N}$)" means that $A = k \times B$, $k \in \mathbb{N}$ not k is a polynomial. $\rightarrow$ some students write it as "$-b^k + p(a - b)$" or "$a^k - p(a - b)$" Some students do not prove the case when $n=k+1$ by using the statement from $n=k$, which is against the method of mathematical induction.